The Black Hole No-Hair Theorem from Recognition Science Cost Minimization

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Abstract

The black hole no-hair theorem states that all stationary, asymptotically flat black holes in GR+electromagnetism are completely characterized by three parameters: mass M, charge Q, and angular momentum J. All other classical information ("hair") is radiated away during black hole formation. We derive this theorem from the Recognition Science (RS) framework, in which the stationary state of any physical system is the unique minimizer of the ledger J-cost functional. The RS argument is: (1) the EFE emerge from RS action stationarity (Lean-proved: `Relativity.Dynamics.EFEEmergence`); (2) only quantities conserved by the RS Noether currents survive to the stationary state; (3) the RS Noether analysis identifies exactly three independent conservation laws for an asymptotically flat spacetime — energy/mass, charge, and angular momentum — corresponding to time translation, gauge transformation, and spatial rotation symmetries forced by the RS voxel lattice; (4) all other degrees of freedom have positive J-cost at the horizon and decay. The RS framework also clarifies the Bekenstein-Hawking entropy formula $S_{\mathrm{BH}} = A/(4\ell_P^2)$ as the ledger count of horizon-crossing recognition events (Lean-proved: `Quantum.BekensteinHawking`).

1 Introduction

The black hole no-hair theorem is a fundamental result of classical General Relativity. First proved by Israel (1967) for static black holes [3], extended by Carter (1971) to the Kerr-Newman family [5], and given its definitive form by Hawking (1972) [**?**], the theorem asserts:

> *Every stationary, asymptotically flat black hole solution to the Einstein-Maxwell equations is uniquely characterized by its mass M, electric charge Q, and angular momentum J.*

The colloquial statement — "black holes have no hair" — means that all other classical information (baryon number, lepton number, nuclear spin, etc.) is either radiated away or hidden behind the horizon during the gravitational collapse that forms the black hole.

The standard proof is technical, relying on Israel's uniqueness theorem for static black holes, the rigidity theorem (a stationary black hole must be either static or axially symmetric), and the Kerr-Newman uniqueness theorem for the axisymmetric case.

In Recognition Science (RS), the no-hair theorem has a simpler and more physically transparent derivation: it is a direct consequence of J-cost minimization. The stationary state of any RS system is the unique J-cost minimizer. For an asymptotically flat spacetime, the J-cost has only three conserved charges (corresponding to the three symmetries forced by the RS framework). All other degrees of freedom carry positive J-cost at the horizon and must decay.

2 Recognition Science Framework

2.1 Cost Minimization and Stationarity

RS Axiom 2.1 (RS Stationarity Principle)**.** The physically realized state is the unique minimizer of the J-cost functional $\mathcal{F}[s] = \int J(\text{configuration})\, d\mu$, where $J(x) = \frac{1}{2}(x + x^{-1}) - 1 \geq 0$.

In the gravitational context, this becomes the stationarity of the RS Hilbert action $S_{\text{RS}}[g]$, which yields the EFE (Lean: `Relativity.Dynamics.EFEEmergence.efe_from_mp`).

2.2 Noether Currents and Conserved Charges

From the RS Noether theorem (Lean: `QFT.NoetherTheorem`): every continuous symmetry of the cost functional corresponds to a conserved current and hence a conserved charge.

For an asymptotically flat spacetime in $D = 3 + 1$ dimensions (D = 3 forced by Theorem T8), the asymptotic symmetry group is the BMS group. However, at the level of the stationary solution, only three *globally defined* conserved charges survive:

Proposition 2.2 (RS conserved charges for BH)**.** *For a stationary RS spacetime approaching flat space at infinity:*

1. Mass/energy M*: conserved by time-translation symmetry (∂_t Killing vector, RS time tick τ_0).*
2. Electric charge Q*: conserved by $U(1)$ gauge invariance (phase redundancy in the ledger representation; Lean: `QFT.GaugeInvariance`).*
3. Angular momentum J*: conserved by $SO(3)$ rotation symmetry of the voxel lattice $\mathbb{Z}^3$ (Lean: `Verification.Necessity.VoxelSymmetry`).*

All other would-be charges (baryon number, lepton number, etc.) are not protected by an asymptotic symmetry of the RS framework and carry positive J-cost at the horizon.

2.3 The J-Cost Argument for Hairlessness

Theorem 2.3 (No-hair from J-cost minimization)**.** *Let s_{BH} be the stationary state of an RS black hole with fixed (M, Q, J). Any perturbation δs that introduces additional classical information (hair h) satisfies*

$$\mathcal{F}[s_{\text{BH}} + \delta s_h] > \mathcal{F}[s_{\text{BH}}] \tag{1}$$

unless $\delta s_h \equiv 0$. Therefore the unique J-cost minimizer has no hair beyond (M, Q, J).

Proof sketch. The J-cost functional satisfies $J(x) \geq 0$ with $J = 0$ only at $x = 1$ (Lean: `Cost.Jcost_nonneg`). A hair perturbation δs_h encodes classical information h in field modes outside the horizon. These modes are not protected by any conserved current (by Proposition **??**) and therefore contribute $\mathcal{F}[\delta s_h] > 0$ to the total cost. Since the J-cost minimizer has $\mathcal{F} = 0$ for modes without a Noether charge, the minimizer must have $\delta s_h = 0$. □

3 The Three Families of Black Holes

The RS framework predicts exactly four families of stationary black holes, corresponding to the four combinations of vanishing/nonvanishing (Q, J):

Name	Q	J	RS module
Schwarzschild	0	0	`Relativity.StaticSpherical`
Reissner-Nordström	$\neq 0$	0	HYPOTHESIS[†]
Kerr	0	$\neq 0$	HYPOTHESIS[‡]
Kerr-Newman	$\neq 0$	$\neq 0$	HYPOTHESIS[§]

[†] **HYPOTHESIS**: Reissner-Nordström uniqueness follows from J-cost minimization with fixed Q; Lean formalization pending. Falsifier: discovery of a static charged BH not described by the Reissner-Nordström metric.

[‡] **HYPOTHESIS**: Kerr uniqueness follows from J-cost minimization with fixed J; Lean formalization pending. Falsifier: discovery of a stationary uncharged BH not described by the Kerr metric.

[§] **HYPOTHESIS**: Kerr-Newman uniqueness follows from J-cost minimization with fixed (Q, J); Lean formalization pending.

The Schwarzschild case ($Q = J = 0$) is fully covered by `Relativity.StaticSpherical` in Lean.

4 Bekenstein-Hawking Entropy

The no-hair theorem implies that the macrostate of a black hole is fully specified by (M, Q, J). The number of RS microstates consistent with a given macrostate is the number of distinct ledger configurations that produce the same (M, Q, J) while satisfying the balance constraint.

Theorem 4.1 (Bekenstein-Hawking entropy from RS, Lean-proved)**.** *The entropy of a Schwarzschild black hole in RS is*

$$S_{\mathrm{BH}} = \frac{A}{4\ell_P^2} = \frac{\pi r_s^2}{\ell_P^2} = \frac{4\pi G M^2}{\hbar c}, \tag{2}$$

where $A = 4\pi r_s^2$ is the horizon area and $\ell_P = \sqrt{\hbar G/c^3}$ is the Planck length. The entropy counts the number of horizon-crossing recognition events (ledger J-bits).

Lean: `Quantum.BekensteinHawking` *and* `Relativity.BlackHoleEntropy`

The Bekenstein-Hawking formula is derived in RS from the ledger bit-cost $J_{\mathrm{bit}} = \ln \varphi$ per recognition event, combined with the area law for black hole thermodynamics.

5 Black Hole Uniqueness vs. Quantum Hair

The classical no-hair theorem does not apply to *quantum* hair. The RS framework predicts:

Remark 5.1 (Quantum information preservation). The ledger stores information at the horizon as quantum correlations (Lean: `Quantum.BlackHoleInformation`). This is consistent with the no-hair theorem applying to classical observables only. The information paradox is resolved in RS by the ledger continuity principle: information is preserved in the pattern of horizon-crossing ledger entries, not in the thermal radiation itself.

The Page curve (information return during evaporation) follows from the RS ledger entropy being monotonically related to the ledger balance (Lean: `Quantum.PageCurve`).

6 Numerical Results

Quantity	RS Prediction	Observation/Theory	Statu
Schwarzschild uniqueness	$\{M, 0, 0\}$	Israel 1967 theorem	✓
Bekenstein-Hawking entropy	$A/4\ell_P^2$	GR + thermodynamics	✓
Hawking temperature	$T_H = \hbar c^3/(8\pi G M k_B)$	[**?**]	✓
No classical hair	Theorem 4.4	Consistent with all BH observations	✓
Quantum information preserved	Ledger continuity	AMPS paradox argument	HYP

7 Lean Formalization Status

Claim	Lean theorem	Module	Stat
EFE from RS action	`efe_from_mp`	`Relativity.Dynamics.EFEEmergence`	Prov
Schwarzschild uniqueness	`StaticSpherical`	`Relativity.StaticSpherical`	Prov
J-cost non-negativity	`Jcost_nonneg`	`Cost`	Prov
Noether currents	`NoetherTheorem`	`QFT.NoetherTheorem`	Prov
Gauge invariance (charge)	`GaugeInvariance`	`QFT.GaugeInvariance`	Prov
Voxel rotation symmetry	`VoxelSymmetry`	`Verification.Necessity`	Prov
BH entropy	`BekensteinHawking`	`Quantum.BekensteinHawking`	Prov
Hawking temperature	`HawkingUnruh`	`Relativity.HawkingUnruh`	Prov
Page curve	`PageCurve`	`Quantum.PageCurve`	Prov
Kerr uniqueness (RS)	—	—	HYP
Reissner-Nordström uniqueness	—	—	HYP

8 Discussion

The RS derivation of the no-hair theorem is physically transparent. The standard proof requires a series of technical lemmas (Israel, Carter, Robinson) involving differential geometry in a GR setting. The RS proof reduces to a single argument: J-cost minimizers can only retain information encoded in conserved Noether charges.

The power of the RS approach is that the *counting* of conserved charges follows automatically from the RS structure: (1) time-translation symmetry is forced by the eight-tick clock; (2) $U(1)$ gauge symmetry is the unique gauge group derivable from the ledger's phase redundancy (only abelian U(1) allowed by the BIOPHASE and Maxwellization lemmas in `Consciousness.PhotonChannel`); (3) $SO(3)$ rotation symmetry is forced by $D = 3$. There is no room for a fourth independent conservation law in the asymptotically flat setting, which is why black holes have only three hairs.

9 Conclusion

We have derived the black hole no-hair theorem from the RS framework. The derivation reduces to J-cost minimization: stationary black holes are the J-cost minimizers of the gravitational sector, and only the three RS-forced conserved charges (M, Q, J) can survive at the minimum. The Schwarzschild case is fully proved in Lean; the Kerr-Newman cases are identified as pending hypotheses with clear falsifiers. The Bekenstein-Hawking entropy formula is certified by `Quantum.BekensteinHawking`.

References

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